

ENGINEERING CHANGE ORDER:

ECO - 3/1/2030:

Requestor: Prof. Kidd

Description of change: Modify your design of PROJECT-CE to completely replace the `milli()` function defined in `arduino.h` with one written by you. This change must include the accompanying register and other constants as well as the ISR. There is no change to the Customer Requirements or the High-level requirements defined in the design document for PROJECT-CE.

Reason for change: Results in

- (1) a more flexible design allowing the design team customize and extend the function,
- (2) a design entirely under company control isolated from any future changes in the 3rd party `arduino.h`, and
- (3) avoids possible future IP ownership claims. Though there is some additional initial cost, we expect that the added independence, flexibility, extensibility and reuse will ultimately save labor and reduce costs in future projects.

DEFINITIONS:

"PROJECT-ASYNC", "PROJECT-CE" - The previous projects upon which this project is based

"`arduino.h`" - helper functions for the Arduino development platform maintained by Arduino

AG, owned by Qualcomm. IP - Intellectual Property

ISR - Interrupt Service Routine

PWM - Pulse Width Modulation

EDUCATIONAL CONSTRAINTS: (changed)

EC.1.0 You must replace the `milli()` function provided by `arduino.h` with your own. Your replacement must be named `mymilli()`. Outside of replacing the functionality, you may extend its capability beyond that of `milli()`. EC.1.0.1 You must write your own PWM counter ISR using it to provide the timing mechanism underlying `mymilli()`. EC.1.0.2 You must define your own register constants that would normally be defined in `avr.h`. These constants do not need to have the same names as those in `avr.h`.

EC.1.1 The system testing and validation requirements must contain those defined in PROJECT-CE. EC.1.1.1 The testing and validation requirements MUST be extended to contain any additional tests necessary to properly evaluate/demonstrate the function of the modified system.

EC.1.2 The final report must be a new revision of the formal design document produced in PROJECT-CE. Make sure you note any changes you've made to the design.

EC.1.3 You must provide a properly formatted ECN notifying other teams of the changes